

GRADE 10 • ALGEBRA II • TOPIC 2

Lesson 2-3

Factored Form of a Quadratic Function

Quadratic Functions and Equations

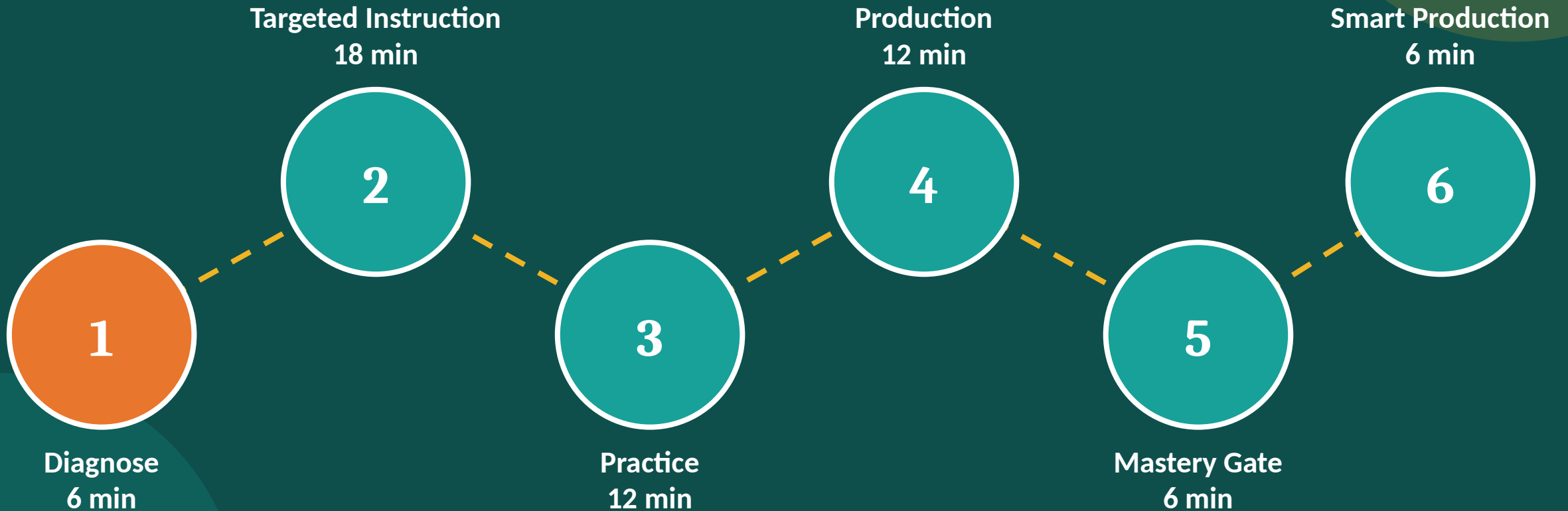
Mr Malek Thiab

$$f(x) = a(x - p)(x - q)$$

Journey of 6 stations, 60 minutes

Today's journey

Six stations. Each one zooms in as we arrive.



What we are learning

OBJECTIVES

1. Solve quadratic equations by factoring.
2. Find the zeros of a quadratic function by factoring.
3. Determine the intervals where a quadratic function is positive or negative.

VOCABULARY

factored form

negative interval

positive interval

x-intercepts

zeros (of a quadratic function)

zero-product property

Standards

HSA.SSE.A.1.B Interpret complicated expressions by viewing one or more of their parts as a single entity.

HSA.SSE.B.3.A Factor a quadratic expression to reveal the zeros of the function it defines.

HSA.APR.B.3 Identify zeros of polynomials when suitable factorizations are available, and use the zeros to construct a rough graph of the function defined by the polynomial.

HSF.IF.C.7 Graph functions expressed symbolically and show key features of the graph, by hand in simple cases and using technology for more complicated cases.

HSF.IF.C.7.A Graph linear and quadratic functions and show intercepts, maxima, and minima.

HSA.SSE.A.2 Use the structure of an expression to identify ways to rewrite it. For example, see $x^4 - y^4$ as $(x^2)^2 - (y^2)^2$, thus recognizing it as a difference of squares that can be factored as $(x^2 - y^2)(x^2 + y^2)$.

HSF.IF.C.8.A Use the process of factoring and completing the square in a quadratic function to show zeros, extreme values, and symmetry of the graph, and interpret these in terms of a context.

STATION 1 • DIAGNOSE • 6 MIN

Start here: three quick checks

1

$$(x + 2)(x - 5) = ?$$

2

Factor: $x^2 + 5x + 6$

3

If $a \cdot b = 0$, then $a = ?$ or $b = ?$

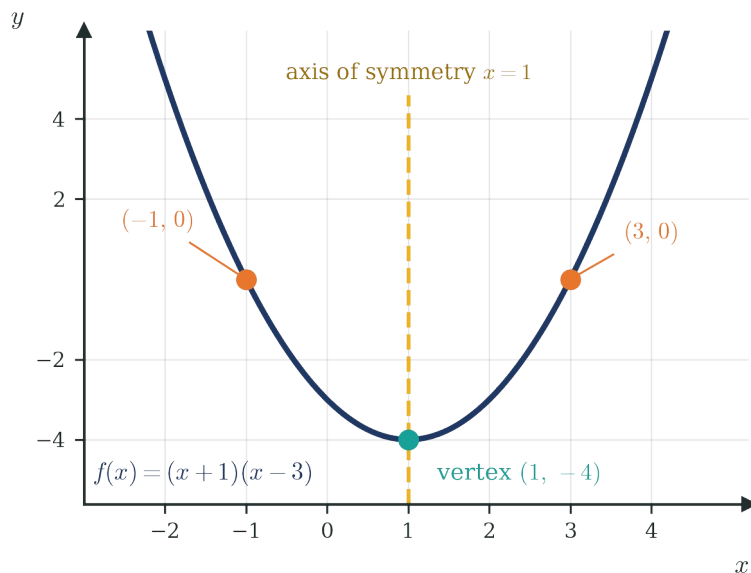
Done when

All three are answered without a calculator, and you can say why a product equal to 0 forces one of its factors to be 0.

STATION 2 • TARGETED INSTRUCTION • 18 MIN

Factored form shows the zeros

$$f(x) = a(x - p)(x - q) \quad \text{zeros: } x = p, x = q \quad \text{axis: } x = \frac{p + q}{2}$$



EXAMPLE A

$$f(x) = (x + 1)(x - 3):$$

find the zeros

$$x + 1 = 0 \Rightarrow x = -1$$

$$x - 3 = 0 \Rightarrow x = 3$$

x-intercepts: $(-1, 0)$, $(3, 0)$

EXAMPLE B

Axis and vertex of the same f

$$x = \frac{-1 + 3}{2} = 1$$

$$f(1) = (2)(-2) = -4$$

vertex $(1, -4)$

STATION 2 • TARGETED INSTRUCTION • 18 MIN

Solve by factoring

$a \cdot b = 0 \Rightarrow a = 0$ or $b = 0$ Solutions of $f(x) = 0$ are the zeros of f

EXAMPLE A

Solve: $x^2 - 7x + 12 = 0$

$$(x - 3)(x - 4) = 0$$

$$x - 3 = 0 \text{ or } x - 4 = 0$$

$$x = 3 \text{ or } x = 4$$

EXAMPLE B

Solve: $2x^2 - 2x - 12 = 0$

$$2(x^2 - x - 6) = 0$$

$$2(x - 3)(x + 2) = 0$$

$$x = 3 \text{ or } x = -2$$

EXAMPLE C

Solve: $x^2 + 3x = 10$

$$x^2 + 3x - 10 = 0$$

$$(x + 5)(x - 2) = 0$$

$$x = -5 \text{ or } x = 2$$

STATION 2 • TARGETED INSTRUCTION • 18 MIN

Positive and negative intervals

$f(x) > 0$ above the axis, $f(x) < 0$ below Zeros split the x -axis; test one x in each part

EXAMPLE A

$$f(x) = (x + 1)(x - 3)$$

Zeros: $x = -1, x = 3$

$$f(0) = -3 < 0 : f < 0 \text{ on } -1 < x < 3$$

$$f > 0 \text{ on } x < -1 \text{ or } x > 3$$

EXAMPLE B

$$f(x) = -(x - 1)(x + 4)$$

Zeros: $x = 1, x = -4$

$$f(0) = 4 > 0 : f > 0 \text{ on } -4 < x < 1$$

$$f < 0 \text{ on } x < -4 \text{ or } x > 1$$

EXAMPLE C

$$f(x) = x^2 - 6x + 8$$

$$f(x) = (x - 2)(x - 4)$$

$$f(3) = -1 < 0 : f < 0 \text{ on } 2 < x < 4$$

$$f > 0 \text{ on } x < 2 \text{ or } x > 4$$

STATION 3 • PRACTICE • 12 MIN

Give it a go: Practice, Apply, Investigate

PRACTICE

Secure the method

Solve: $x^2 + 8x + 15 = 0$

Zeros of $f(x) = (x + 3)(x - 2)$

Done when: each solution is checked by substitution and the zeros are also written as x-intercepts.

APPLY

Use it in context

Write f with zeros -2 , 4 and $f(0) = -16$

Where is $f(x) = x^2 - x - 6$ negative?

Done when: your function is checked by substituting a point and the interval is written with strict inequality signs.

INVESTIGATE

Find out why

Show the axis of $a(x - p)(x - q)$ is $x = \frac{p + q}{2}$

Why is
 $(x - 2)(x - 3) = 6 \Rightarrow x - 2 = 6$ wrong?

Done when: your argument uses the zero-product property or the symmetry of the zeros, not just a computed answer.

STATION 4 • PRODUCTION • 12 MIN

Production: the Red Sea dive-tour profit

THE MODEL

A Red Sea tourism operator off the Jeddah coast models its daily profit from a dive-boat tour as a quadratic function of the ticket price, in the spirit of Vision 2030 tourism growth. All numbers are model values invented for this lesson, not real company data.

$$P(x) = -(x - 20)(x - 140)$$

P and x both in SAR

1. Find the two break-even ticket prices (the zeros) and write them as x-intercepts.
2. Find the prices where the tour makes a profit and the price that gives the largest profit.
3. Find the profit at a ticket price of 100 SAR and compare it with the largest profit.

DONE WHEN

- Both zeros are found by setting each bracket equal to 0 and are labelled with SAR.
- The profit interval is written with strict inequalities and explained in a sentence.
- The best price comes from the axis of symmetry and is checked by substitution.
- Every answer has units (SAR) and a sentence about what it means for the operator.

Hint for solving: the best price is halfway between the break-even prices; substitute it into the function to get the largest profit.

STATION 5 • MASTERY GATE • 6 MIN

Mastery Gate: three questions to pass

1

Zeros and axis of $f(x) = (x + 6)(x - 2)$

2

Solve: $x^2 - x - 12 = 0$

3

Where is $f(x) = (x + 2)(x - 5)$ negative?

Pass mark

3 of 3 correct unlocks Station 6. Fewer than that: return to the slide for the question you missed, redo it, then re-take the gate.

STATION 6 • SMART PRODUCTION • 6 MIN

Smart Production: make one, then check it

CREATE

1. Write one quadratic function in factored form with two integer zeros.
2. Write the zeros, the axis, the vertex and the intervals where it is positive and negative on the back.
3. Swap with a partner, solve theirs and check it by substituting a point.

Done when: your solution is complete and correct, and your partner agrees with every step.

TIME TO CHECK

Exit ticket: write your answer on a sticky note and place it on the door.

$$\text{Solve } x^2 - 6x + 5 = 0$$

$$\text{Where is } x^2 - 6x + 5 < 0?$$

Next lesson: 2-4 Overview of Complex Numbers.

Exam connections

SAT

Parabola with x -intercepts -3 and 5

passes through $(1, -32)$; find a

STEPS

$$1 \quad y = a(x + 3)(x - 5)$$

$$2 \quad -32 = a(4)(-4) = -16a$$

$$3 \quad a = 2$$

GAT

A = larger solution
of $x^2 - x - 6 = 0$

$$B = 3$$

STEPS

$$1 \quad (x - 3)(x + 2) = 0$$

$$2 \quad x = 3 \text{ or } x = -2$$

$$3 \quad A = 3 = B$$

SAAT

Sum of the solutions
of $x^2 - 5x + 6 = 0$

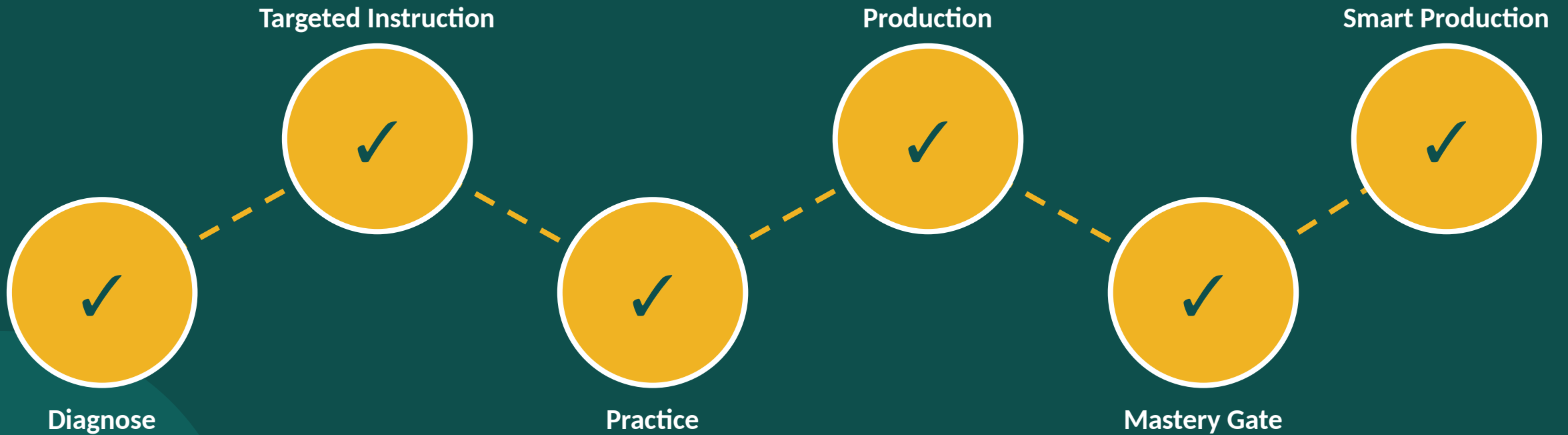
STEPS

$$1 \quad (x - 2)(x - 3) = 0$$

$$2 \quad x = 2 \text{ or } x = 3$$

$$3 \quad 2 + 3 = 5$$

Journey complete



Factor, set each bracket to 0, and read the zeros, the axis and the intervals where a quadratic is positive or negative.